

Autocatalytic Closure and Chemical Memory: When Reaction Networks Begin to Remember

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Abstract

This paper isolates autocatalytic closure from later claims about alphabets, homochirality, or protocells. A reaction network begins to remember only when its products help regenerate the conditions that make them. In CDFD terms, this is the first serious biological use of retained structural memory M_s : retained organization through feedback. The companion script performs a conservative closure scan and reports sub-ignition persistence enhancement rather than completed metabolic ignition. The qualitative closure criterion developed here (§2.4) is a mean-field caricature of a more rigorous existing formalism — reflexively autocatalytic, food-generated (RAF) set theory (Hordijk and Steel, 2004) — which future revisions of this paper should engage directly rather than reinvent independently.

Keywords: autocatalysis; chemical memory; reaction networks; origins of life; computational modelling.

Preprint note

This is a computational research article. The companion script produces numerical diagnostics reported in the Computational Results section. It is not a narrative review and does not claim historical proof of a unique abiogenesis pathway.

Symbol Conventions

See Paper 1, §Symbol Conventions, for the base CDFD and Greek-symbol tables. This paper introduces species-specific transport resistance C_i (not to be confused with product concentration symbols in legacy drafts) and the ignition parameter Γ (§2.3).

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1 Introduction

Papers 1–5 supplied energy, transport, compartments, and selective interfaces. None of these alone guarantees *self-reinforcing* chemistry. Autocatalysis and cross-catalysis supply positive feedback: products accelerate their own formation [1, 3]. In open systems, however, diffusion and dilution supply negative feedback. Closure is the balance point.

A remaining question is how catalytic selection escalates into network persistence.

Capstone channel mapping. Metabolic closure is not single-channel: losing input Φ , electron throughput σ_e , proton coupling σ_p , or stabilization capacity against S collapses the bundled J in the weakest-link sense even when Γ inequalities locally favor amplification.

2 Model and Assumptions

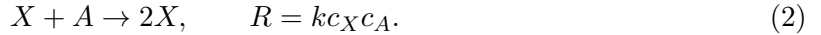
2.1 Network mass balances

For species X_i with concentrations c_i ,

$$\frac{\partial c_i}{\partial t} = \sum_j R_{ij}(\mathbf{c}) - \nabla \cdot \mathbf{J}_i + S_i, \quad \mathbf{J}_i = -\frac{1}{C_i} \nabla c_i, \quad (1)$$

with sources S_i . Here C_i is a *transport resistance* for species i (not to be confused with product concentration symbols in legacy drafts).

2.2 Minimal autocatalytic core



Well-mixed loss to a compartment of scale L scales as $D_{\text{eff}} c_X / L^2$ with $D_{\text{eff}} \sim 1/C$ in the resistance picture. Then

$$\frac{dc_X}{dt} = kc_X c_A - \frac{1}{CL^2} c_X = c_X \left(kc_A - \frac{1}{CL^2} \right). \quad (3)$$

2.3 Ignition parameter

Define

$$\Gamma = kc_A CL^2. \quad (4)$$

If $\Gamma > 1$, exponential growth of c_X results in this linearized well-mixed limit; if $\Gamma < 1$, trajectories decay toward zero without other sources. The boundary $\Gamma = 1$ is a mean-field caricature; spatial patterning and nonlinear saturation modify bifurcations.

2.4 Vector closure

A network is *closed* if every required intermediate is produced internally faster than it leaks, $\forall i, \sum_j k_{ij} c_j > (C_i L_i^2)^{-1}$ in the same schematic sense.

This criterion is a linearized mean-field approximation of the network-closure question, and the more rigorous existing formalism for this exact problem is RAF (reflexively autocatalytic, food-generated) set theory [2], which formalizes closure via a directed hypergraph over catalyzed reactions and a food set, without requiring the linearization used here. Future work on this paper should test whether the Γ threshold criterion and RAF-set membership agree on the same toy networks, since the two currently make the same qualitative claim through different mathematical machinery without having been reconciled.

2.5 Energy coupling

Rates may scale with coupled fluxes $R_i \propto |\mathbf{J}_e \cdot \mathbf{J}_p|$ when redox drive gates chemistry [4]. We do not specify constants.

2.6 Linear stability (Jacobian)

Let $\mathbf{J}_{\text{jac}}$ be the Jacobian of the reaction vector field. Local stability of a steady state requires $\text{Re } \lambda < 0$ for all eigenvalues λ of $\mathbf{J}_{\text{jac}}$. Autocatalysis can instead produce unstable modes, limit cycles, or blow-up depending on constraints and saturation.

3 Main Results

3.1 Conceptual

Metabolic organization begins when positive feedback loops beat transport losses [5]. Compartments increase C (resistance to escape), improving Γ for fixed k, c_A, L .

4 Numerical Verification

`supplementary_o01_06.py` integrates a reduced two-parameter caricature. Reported final Φ values:

- $(\Phi_0 = 0.1, C = 1.0) \rightarrow 0.073$
- $(\Phi_0 = 0.5, C = 1.0) \rightarrow 0.554$
- $(\Phi_0 = 0.1, C = 5.0) \rightarrow 0.102$
- $(\Phi_0 = 0.5, C = 5.0) \rightarrow 0.849$

Higher constraint C raises the final state relative to comparable baselines, but **no run exhibits runaway ignition** in the companion script. Narratives about “critical metabolism” must therefore remain Hypothesis/Open until bifurcation scans identify supercritical parameters.

5 Interpretation and Limitations

The mathematics clarifies what must be proven: closure is a competition between amplification and loss. The numerics currently demonstrate *sub-ignition* persistence enhancement, not autonomous self-acceleration to a living steady state.

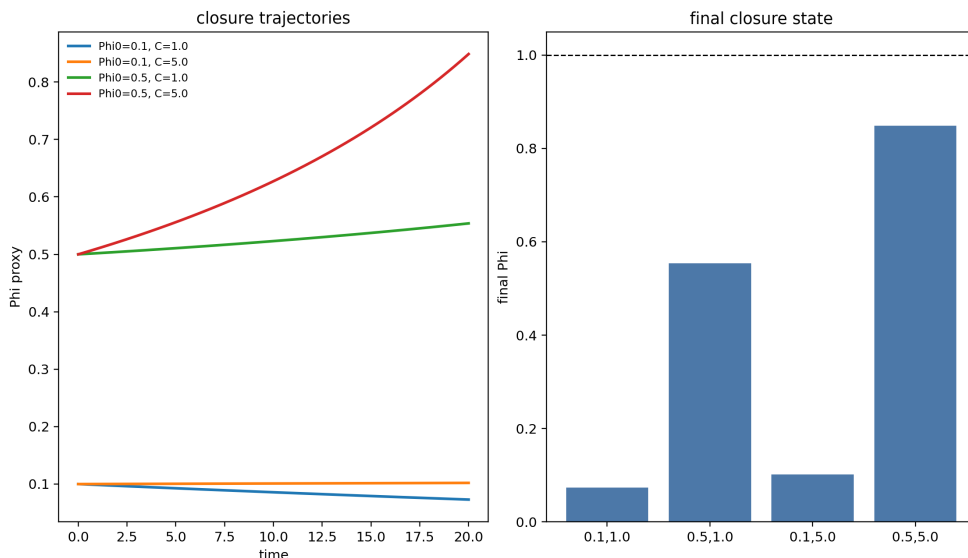


Figure 1: Autocatalytic-closure diagnostic. The trajectories show persistence enhancement under stronger constraint but do not prove completed metabolic ignition. The figure is a disciplined threshold check, not a claim of living metabolism.

- Mismatch between rhetorical “ignition” and outputs is explicitly documented.
- Need bifurcation analysis in (k, c_A, C, L) space with spatial resolution.
- Saturation chemistry and side reactions omitted.
- The RAF-set reconciliation flagged in §2.4 has not been performed.

6 Modern Network Boundary

Autocatalysis is not enough by itself. A closure claim must distinguish three layers: self-amplification of a species, mutual regeneration of a reaction set, and persistence of an organized network through environmental cycling. The strict reading is that closure becomes biological only when it couples to chemical memory, boundary retention, and later copying. Homochirality and genetic inheritance are treated separately (Papers 7 and 9).

7 Conclusion

This paper states the closure criterion clearly and shows that the companion script does not yet realize full ignition. Aromatic carriers, where pattern persistence becomes information-relevant, are treated in Paper 7;

8 Reproducibility Note

python supplementary_o01_06.py

Related literature

This manuscript is self-contained within the Part II origins-of-life sequence. Shared CDFD/CDFL symbols (Φ , C , S , M_s , Ψ_s) are used as bookkeeping only; they do not require Part I physics results. Field anchors are cited in the body and bibliography.

Competing Interests

The author declares no competing interests.

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Data and Code Availability

Companion scripts, generated figures, and numerical outputs for this manuscript are included in the public CDFD Part II release archive (<https://github.com/bampita-bico/CDFD-Part-II-Release>; Zenodo DOI 10.5281/zenodo.20264779). See the Reproducibility Note above for the paper-local script path.

Author Contributions

Conceptualization, formal analysis, computational diagnostics, and manuscript preparation: Steve Bico Mujjabi

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